Obsidian procurement and consumption on Easter Island

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Introduction

A preliminary investigation of the Orito obsidian quarry on Easter Island, Chile was conducted in April of 1981. The goals of this research were to determine the lithic reduction strategies employed and the period over which the quarry was in use. Although previous archaeological research has examined lithic assemblages (McCoy 1976, Mulloy 1961, Smith 1961), it has focused on problems of artifact typology or minor aspects of production. This paper establishes a preliminary model of obsidian reduction at the quarry which may be related to an island-wide characterization of site assemblages in behavioural terms. In addition, the relationship between obsidian mining and consumption is examined over time.

Easter Island is a good location for the study of temporal change in lithic technology since no technological diffusion appears to have occurred after initial settlement. The obsidian on Easter Island is abundant and there appears to be no decrease in superior material or quantity through time. Territorial control over the three obsidian flows may have been a factor during the later parts of prehistory since there is documentation—both historical and archaeological—of conflicts arising near the end of the Expansion Phase (AD 1000-1680) (Metraux 1940). These social constraints and the limited distribution of obsidian sources (Figure 1) may have played an important role in determining use and production at the Orito quarry. Obsidian hydration dating of samples from open-pit mines was begun to help our considerations of these questions.

Orito quarry

The Orito quarry is one of three obsidian outcrops located on the southwestern corner of Easter Island (Figure 1). The site consists of shallow depressions or open-pit mines distributed across the slopes. A large volume of surface debitage shows the extensive prehistoric use of obsidian for tool manufacture. The site has changed little since stone tool production ceased although trampling by people and grazing animals has slightly modified the surface debris.

The obsidian was quarried prehistorically by digging into the side of the hill. The resulting pits tended to be circular to oval with the long axis running across the slope of the hill (Figure 2). No pits were found which exceeded a metre in depth, with the majority shallower. The obsidian is commonly found as slabs with the larger pieces being about 20-30 cm long. The slabs are usually only about 10 cm thick.

The study area is located on the north slope of Mt Orito, but quarrying activity was noted around the circumference of the hill. The area surveyed for quarrying pits included the entire slope of the hill. Most pits occur in the midslope area of the hill. A total of 33 pits were mapped.

A single mining pit was excavated to determine the general quarrying technique and to collect debris for an evaluation of the reduction strategies employed at the site. A 7 × 1 m trench went through the centre of Pit 1, down the slope of the hill. A surface collection of the trench was made in one metre segments. The test unit was then excavated in 10 cm levels. The contours of the surface and the test pit profile indicate that a shallow pit had been excavated, with the majority of the resulting debirs thrown or pushed downslope to create a low ridge. This is the same surface configuration as at the other mining pits.

The lithic debitage at the mine indicates a wide range of tool manufacture at the quarry. All the flaking was done by percussion and hammerstones are found among the mining debris. Flakes modified for use, including the *mataa*, a stemmed obsidian flake used as a spearhead, were common on the surface of the

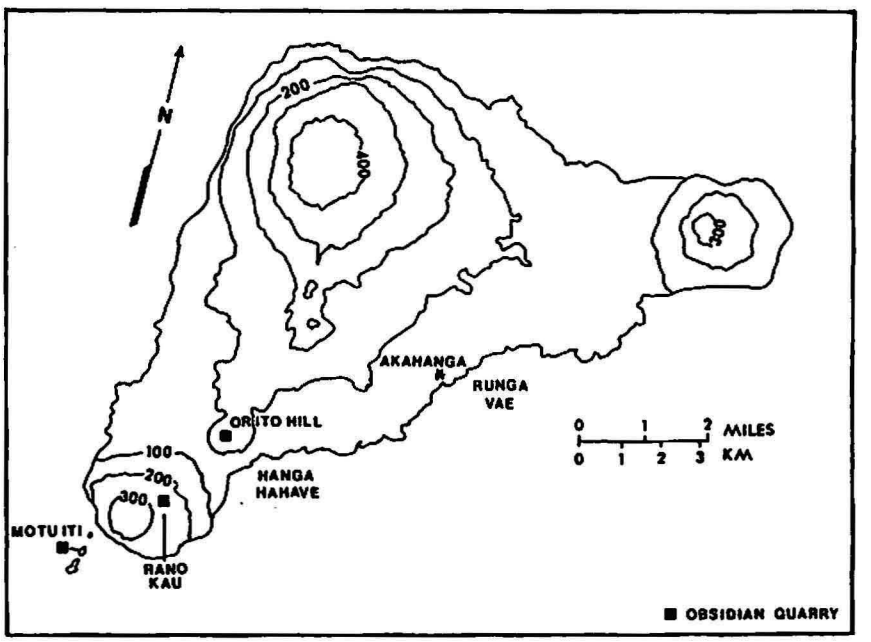

Figure 1. Easter Island.

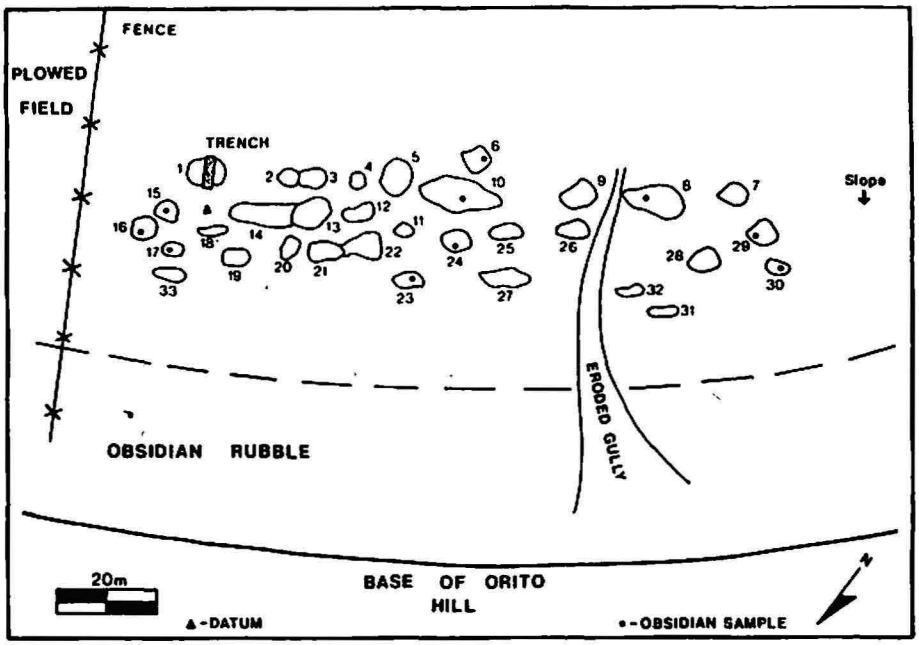

Figure 2. Orito quarry open-pit mines (northeast area).

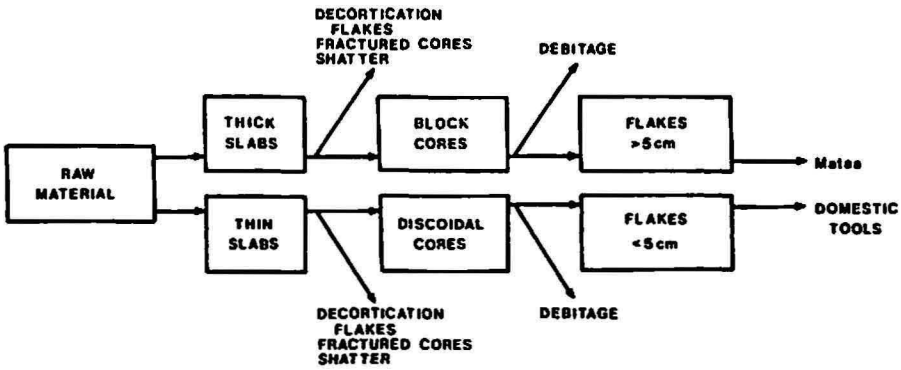

Figure 3. The lithic reduction sequence.

quarry. The test excavation of Pit 1 resulted in the recovery of 6 *mataa*, 2 hammerstones, 11 block cores, and 7 discoidal cores.

Lithic reduction sequence

The lithic sample collected from Orito quarry was analysed to determine the method and extent of lithic tool production at the site. The reduction sequence from raw material to finished tool is presented in Figure 3. A small sample of lithic material from the excavation of a habitation area near *Ahu* Akahanga on the south coast (Fig. 1) has been analysed for comparative purposes. It is used to develop hypotheses concerning differences in reduction activities between quarry and habitation localities.

The natural dimensions of the raw material imposed a significant constraint on the reduction strategies employed for the production of the different tool types. The majority of obsidian at Orito quarry is in the form of tabular slabs which are generally less than 10 cm thick. Two types of cores were produced to accomodate the dimensions of the raw material: discoidal cores (Crabtree 1972) and block cores. A block core is a slab with flakes removed from one or two edges, but only from one face. This core type was best suited for producing large, broad flakes which were needed for the manufacture of *mataa*.

The discoidal cores have flakes removed from both faces and generally from all directions so a circular core with a lenticular cross-section was produced. More flakes, and thus a greater length of cutting edge, could be produced from a discoidal core strategy given the limitations imposed by the size of the raw material. This activity appears to have occurred predominantly at the quarry site. We contend that the discoidal cores were prepared at the quarry site, then transported to habitation sites where flakes were produced when needed. This divergence between quarry production and habitation loci production seems to be evident at the core level and is supported by the data presented below.

The discoidal cores recovered from Easter Island sites are generally circular and tend to be bifacially worked. The thin slabs of obsidian would have been suitable for this type of core although the thicker pieces could also have been used. Medium to small flakes were produced from these cores and represent the majority of the flake tool assemblage recovered from habitation locations.

Discoidal cores were probably prepared at the quarry and then transported to the village area. The lithic artifact assemblage on Easter Island is composed primarily of flake tools. These consist of flakes which were modified slightly to prepare a usable edge, or flakes which were utilized without any edge modification. The relatively low frequency of primary and secondary flakes, especially those larger than 5 cm, from the habitation area studied supports the hypothesis of preliminary core shaping at the quarry locality (Fig. 4). The analysed debitage sample from Akahanga shows a high percentage (75%) of tertiary flakes. Discoidal cores may also have been used as choppers (McCoy 1976).

The block core is a unifacial core which can be either unidirectional or bidirectional. We propose a selection of the thicker slabs of obsidian for this core type. The platform, if it is prepared, is usually altered solely by the removal of the cortex and only a small percentage of the flakes exhibit ground platforms. The decortication of only one surface may have been employed to retain mass so that large broad flakes could be removed. The *mataa* were manufactured on large flakes which were bifacially shaped on the platform end to provide a narrow stem for hafting. The feather-edge was left unmodified and served as the cutting edge. Often the original flake platform remains at the base of the stem; a small percentage retain cortex on the platform. Six completed *mataa* were recovered from the test trench, and a large number

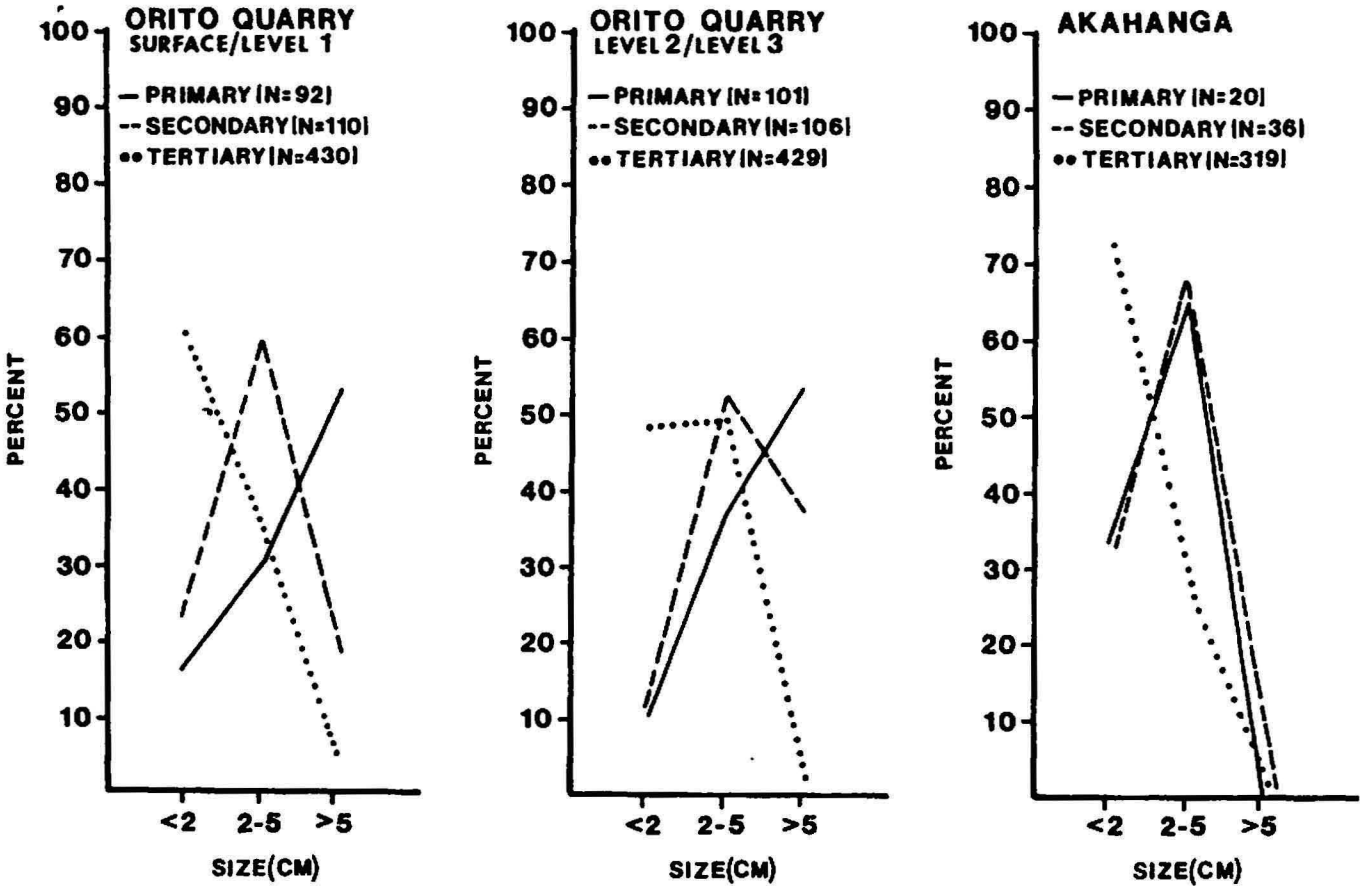

Figure 4. Distribution of flake categories by size from Orito quarry and the Akahanga habitation area.

were observed among the quarry surface debris. The presence of these tools in the final stages of manufacture supports the contention that much of the *mataa* production occurred at the quarry locality. Logistically, *mataa* manufacture at the quarry would be preferable to transporting large obsidian nodules to the sites. Because of the limited core dimensions imposed by the raw material size, it is speculated that only one or two flakes of suitable size for *mataa* production could be produced from each block core. The type of flakes selected for *mataa* production are quite specific and there would have been a large amount of discarded material associated with this activity.

Debitage analysis

The sample recovered from the test trench contains 1268 flakes with striking platforms and 3171 fragments without striking platforms. The flake debitage from the quarry test excavation was analysed by size and type of flake. Only flakes with striking platforms were used in the analysis. The flake types are characterized by the presence and/or coverage of cortex on the flake's platform and dorsal surfaces. The three types are defined as:

a) Primary: cortex on the dorsal surface and platform;

b) Secondary: cortex on a portion of the dorsal surface and cortex on the platform;

c) Tertiary: no cortex remaining on the flake.

The flakes of each type were separated into three size categories, based on the maximum diameter of the flake. The three categories are: smaller than 2.5 cm, 2.5-5 cm, and greater than 5 cm.

For this preliminary study, materials from the surface and level 1 were combined, as were materials from levels 2 and 3. A sample from only one pit mine did not allow for an interpretation of stratigraphic variation but the distinction was made during excavation so that a comparison with future quarry excavations would be possible.

The flake type distribution is very similar for the two units (Fig. 4). The fill material in these pits is probably from slumping and downslope movement of chipping debris. The lower levels had a higher frequency of unworked material which was probably left from the time of initial excavation. Most of this material is unsuitable for core production either because of size or internal flaws.

Several observations can be made concerning the flake type and size distributions. The primary flakes tend to increase in frequency with increasing flake size. This may reflect an attempt to remove the cortex by the removal of a limited number of large flakes leaving a non-cortex surface with a minimal number of irregularities. The decortication flakes would generally have been poorly suited for tool use because the edges were composed of a severely hydrated surface exterior. From site contexts, however, a few cortex flakes have been modified to a functional edge.

A second observed pattern is the low frequency of large flakes (>5 cm) of both the secondary and tertiary types (Fig. 4). Although this general flake size distribution might be expected from the reduction of discoidal or block cores, this could support the selection of large flakes for tool manufacture. Large flakes would then be expected as modified flakes at the quarry and at habitation sites. The *mataa* would require flakes of this size but large flakes would be

Mine Number	Lab Number	Micron Value	Date (AD)
6	82-2258	1.70±.1	1523±56
	82-2259	1.70±.1	1523±56
8	82-2261	2.50±.1	988±81
	82-2262	1.84±.1	1444±61
	82-2263	1.99±.1	1352±64
10	82-2264	1.54±.1	1605±51
	82-2265	1.39±.1	1675±46
	82-2266	1.44±.1	1652±47
15	82-2267	1.76±.1	1490±58
	82-2268	1.06±.1	1803±35
	82-2269	2.25±.1	1177±73
16	82-2270	1.81±.1	1461±59
	82-2271	1.56±.1	1595±51
	82-2272	1.48±.1	1634±49
17	82-2273	2.44±.1	1036±80
	82-2274	2.24±.1	1184±73
	82-2275	1.76±.1	1490±58
18	82-2276	1.64±.1	1554±53
	82-2277	1.79±.1	1473±59
	82-2278	2.28±.1	1156±75
	82-2279	2.25±.1	1177±73
23	82-2246	2.21±.1	1206±72
	82-2247	2.19±.1	1220±72
	82-2248	2.41±.1	1059±79
24	82-2249	1.65±.1	1549±54
	82-2250	0.96±.1	1835±45
	82-2251	1.94±.1	1384±64
29	82-2252	2.02±.1	1333±66
	82-2253	2.02±.1	1333±66
	82-2254	1.39±.1	1675±46
30	82-2255	2.15±.1	1247±70
	82-2256	1.87±.1	1426±61
	82-2257	1.79±.1	1473±59

Table 1. Orito quarry obsidian hydration dates.

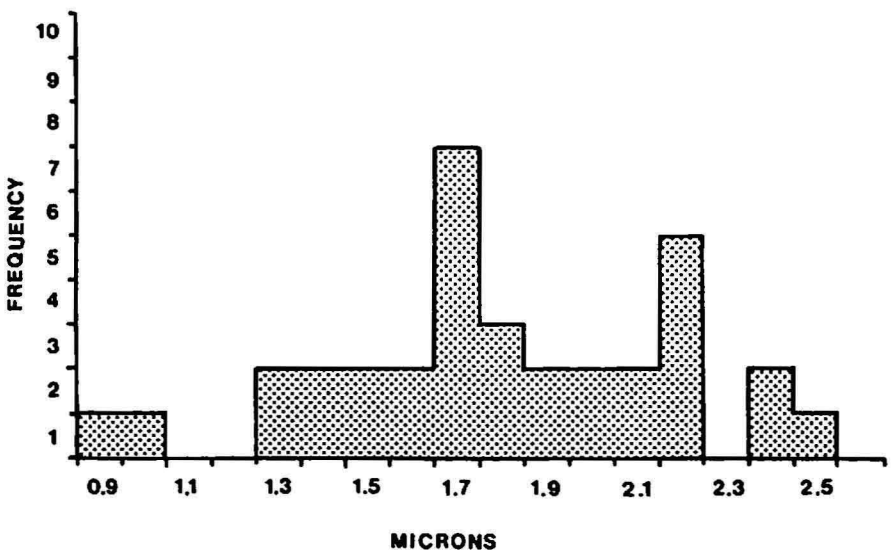

Figure 5. Obsidian hydration dates from Orito quarry.

suitable also for other utilitarian tool types.

If the initial core preparation was done at the quarry, the debitage samples excavated from habitation site contexts should have a low frequency of cortex flakes and a relatively high frequency of small tertiary flakes. Exhausted discoidal cores would also be expected. The small sample of flakes excavated from the *Ahu* Akahanga area did exhibit this pattern (Fig. 4). An analysis of the types of flakes selected for use at the habitation area would help to interpret the function of the discoidal cores.

Temporal use of Orito quarry

From the 33 mines within the north-eastern sector of Orito quarry, 11 were chosen for dating. Small hand-excavated test probes of 25 cm^3 were taken from the centre of these. An average of 3 obsidian specimens per probe was randomly selected for dating, using the obsidian hydration rate of $6.29U^2/Yr^{-1}$(Michels *et al.*, in press). The 33 hydration rim values (Table 1, Fig. 5) indicate an extended period of obsidian mining and reduction at this locality. The earliest hydration rim suggests that mining activity in this section of the quarry began around AD 988 ± 81 (2.50U±.1), and with the exception of two very late dates appears to terminate at AD 1675 ± 46 (1.39U±.1).

Obsidian consumption

The dating of a large sample of residential structures (Stevenson 1984) also permitted the examination of obsidian consumption rates by 50-year phases. It is assumed that the rates of consumption will be directly reflected in discard rates and that the disproportionate deposition by phase is represented in the obsidian samples retrieved from each habitation site. A randomly selected sample of 184 residential sites (66%) from Quadrangles 7 and 11 located inland of Akahanga (Fig. 1) was dated to determine phases of occupation. An average of 9-10 obsidian specimens were dated per site and each obsidian date was assigned to one of eleven 50—year phases. The measure of obsidian consumption for each phase consists of the total frequency of dated obsidian specimens for each 50—year phase divided by the number of separate occupations which occur within each phase (Table 2, Fig. 6). For example, in Phase 6 a total of 157 specimens from 102 different residential sites were dated to between AD 1601 and AD 1650 thus producing an obsidian use ratio of 1.54.

Figure 6 demonstrates that obsidian consumption rises during the first six phases (300 years) until around AD 1650. It then decreases slightly in Phase 7 before rising to a higher consumption rate in Phases 8-10. After AD 1850 it drops dramatically.

This pattern of obsidian use is significantly different from that

Phase no. and AD start date	Number of occupations	Number of Specimens	Consumption Ratio
1 (1350)	42	42	1.00
2 (1401)	63	79	1.25
3 (1451)	71	87	1.22
4 (1501)	93	125	1.34
5 (1551)	101	150	1.48
6 (1601)	102	157	1.54
7 (1651)	90	129	1.43
8 (1701)	113	192	1.70
9 (1751)	112	193	1.72
10 (1801)	108	183	1.69
11 (1851)	40	51	1.21
Totals	935	1388	$\bar{x}$ = 1.42

Table 2. Obsidian consumption ratios per 50-year phase.

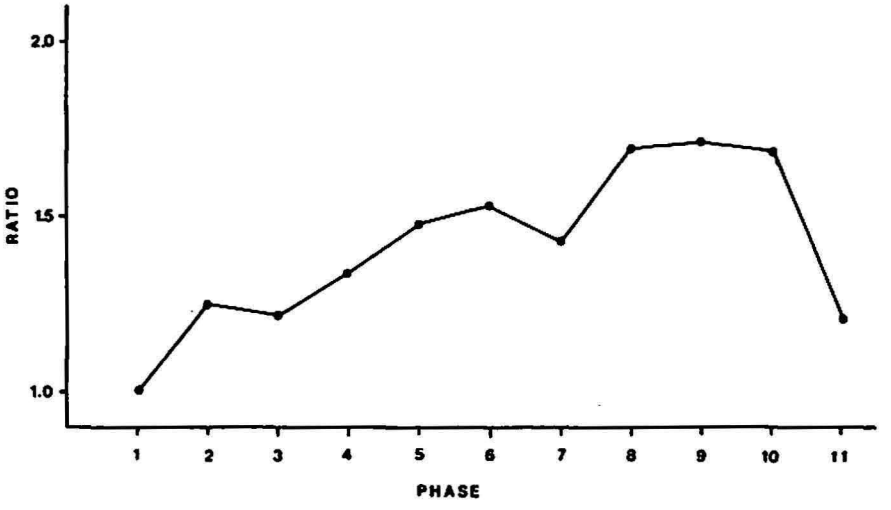

Figure 6. Obsidian consumption curve, AD 1350-1864.

estimated on the basis of obsidian hydration dates from Orito quarry (Fig. 5). The terminal date of AD 1675 ± 46 (Phase 7) from the mines may be misleading. It does suggest that the subsurface extraction of obsidian halted, at least in the northeast section of the quarry. Either open-pit mining continued at other localities on Orito or Rano Kau, or the acquisition of obsidian became less intensive.

Summary and discussion

The preliminary reduction sequence proposed here provides an initial structure for the analysis of habitation site assemblages and will allow the investigation of variation in obsidian reduction and curation. Additional behavioural inferences can be derived from the refinement of the model, and lithic edge damage and microwear studies.

Oral traditions recorded by Metraux (1940) state that, around AD 1680, major transformations occurred in the socio-political structure of the island population. Although the nature of these changes is currently unclear, they are correlated with shifts in religious orientation, interment practices, and the decline in architectural manifestations of differences in rank. The slight decline in population (measured by obsidian consumption) about AD 1650-1700 may reflect a higher death rate or slight population displacement from the central part of the island.

The rise in obsidian consumption during Phases 8 to 10 also corresponds with an increase in population within Quadrangles 7 and 11. This may reflect increased population nucleation in this late phase, a pattern also noticed by McCoy (1976) for the areas inland of Hanga Poukura and Hanga Tee several kilometers to the west. The reasons for the apparent increase in the use of obsidian during these phases is not well understood. An increase in population coupled with the trend toward increasing nucleation may have resulted in the formation of larger residential units and thus an increase in the intensity of domestic activities. An alternate hypothesis is that political and social instability may have resulted in the inability of a specific group to control access to resources, such as obsidian, within its own territory. Thus, greater access to the obsidian quarries may have promoted its use now that costs associated with acquiring the material had been reduced. These, and other hypotheses, await evaluation.

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